higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CONSTRUCTION PLANNING
NQF LEVEL 4

(12010464)

27 February 2019 (Y-Paper)
13:00–16:00

Calculators may be used.

This question paper consists of 8 pages and 1 answer sheet.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.
5. Rule off across the page after each completed question.
6. All work you do NOT want to be marked must be clearly crossed out.

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QUESTION 1: INTRODUCTION TO CONSTRUCTION CONTRACTS

1.1 Define the following terms that are used in construction contracts:

1.1.1 Defects liability period

1.1.2 Interim payments

1.1.3 Tender

1.1.4 Contract sum

1.1.5 Breach of contract

(5 × 2) (10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

1.2.1 A written contract can be produced as evidence in a court of law.

1.2.2 After a contract is awarded, the contract documents are signed only by the contractor.

1.2.3 It is the responsibility of the client to clear the site and do the setting out after a contract was awarded to a contractor.

1.2.4 After completion of the work the client will inspect the project and create a snag list.

1.2.5 The contract must state the date of completion of the project.

(5 × 1) (5)

[15]

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QUESTION 2: BASIC COMMON AND COMMERCIAL LAW APPLICABLE TO CONSTRUCTION

2.1 Choose a description from COLUMN B that matches a term in COLUMN A. Write only the letter (A–F) next to the question number (2.1.1–2.1.6) in the ANSWER BOOK.

COLUMN A		COLUMN B	
2.1.1	Statute law	A	a common-law right of the contractor that comes into effect when the contractor takes possession of the site and starts construction
2.1.2	Builder's lien	B	a law that comes from custom and previous judgements
2.1.3	Common law	C	holding back the sides of an excavation in a situation where it might collapse
2.1.4	Lateral support	D	a term that is not specifically stated and comes from common law and statute law
2.1.5	Implied terms	E	principles and rules laid down in the laws that are written and passed by parliament
2.1.6	Title deed	F	an official document that confirms ownership of a specific property

(6 × 1) (6)

2.2 Having a valid contract or an agreement before commencing with a construction project is very important. It acts as a common understanding between parties and can act as proof during disputes.

Explain the difference between a *contract* and an *agreement*. (2 × 2) (4)

[10]

QUESTION 3: STANDARD CONSTRUCTION CONTRACTS

3.1 Give the full wording of the abbreviation *CIDB*. (1)

3.2 Apart from the CIDB contract document, name any other standard building contract. (1)

3.3 A contractor can participate in a contract in three different ways: as the main contractor, as a subcontractor (a non-nominated/nominated subcontractor) or as a joint venture partner.

Explain how a contractor is selected to participate in a contract as the following:

3.3.1 Main contractor

3.3.2 Subcontractor

3.3.3 A nominated subcontractor

(3 × 2) (6)

3.4 The main contractor has various responsibilities, two of which has to do with:

The design of the building and contract conditions;

List the contractor's responsibility towards the above. (2)

[10]

QUESTION 4: TENDERS AND QUOTES

4.1 A contractor may tender for a project by following the tender process.

Explain what is meant by *tender adjudication process.* (2)

4.2 Name THREE consultants who can assist the client in putting together the tender documents. (3)

4.3 List FIVE documents that are part of the tender documents. (5)

4.4 Explain the importance of the *Bill of quantities* in the project documentation. (3)

4.5 Write the following abbreviations in full:

4.5.1 SARS

4.5.2 VAT

(2 × 1) (2)

[15]

QUESTION 5: QUANTITIES AND COSTING

5.1 Explain your understanding of the term invited tenders. (4)

5.2 A construction company is awarded a contract to build a new warehouse. The dimensions of the warehouse are 12 m × 8 m and a superstructure height of 2,4 m.

NOTE: This warehouse consists of only FOUR external walls and no interior brick walls.

5.2.1 Calculate the total area of the FOUR external walls of the superstructure. (4)

5.2.2 Assume 104 bricks are required in 1 m^2.

Calculate the number of bricks needed to build the external walls of the superstructure. Ignore all openings. (2)

5.2.3 Calculate the total area of all openings (windows/doors) as per the following schedule:

TYPE	NO	WIDTH	HEIGHT
DOOR	1	0.810	2.030
W1	5	1.200	1.200
W2	1	2.000	1.200
W3	1	0.600	0.600

(5)

5.2.4 Calculate the actual number of bricks required to build the superstructure after deduction of all openings (windows/doors). Round off to the next whole brick. (3)

5.2.5 A single brick costs R7,56.

Calculate the total cost of the actual bricks required for the superstructure. (2)

[20]

QUESTION 6: PLANNING AND ARRANGING CONSTRUCTION ACTIVITIES

6.1 A contractor must arrange both critical and non-critical construction activities in a logical and progressive sequence in which each activity follows the other in the correct order.

List FOUR advantages a contractor may obtain by following the correct logical and progressive sequence. (4)

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6.2 The list below shows EIGHT typical activities on a construction site.

Indicate from the list below whether each activity is critical or non-critical.

6.2.1 Fitting of windows and doorframes

6.2.2 Foundation brickwork

6.2.3 Land surveying

6.2.4 Painting of walls

6.2.5 Plastering of walls

6.2.6 Laying of foundation

6.2.7 Fitting ceiling boards

6.2.8 Excavate the foundations

(8 × 1) (8)

6.3 Rearrange the following construction activities in the logical and progressive sequence so that all activities can be completed within the specified timeframe:

- Beam filling
- Building the masonry plinth (foundation) walls
- Removing of all trees, shrubs and vegetation
- Excavation of foundation trenches
- Building of superstructure
- Casting of concrete slab
- Setting out of building
- Pouring of concrete in trenches

(8)

[20]

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QUESTION 7:

The following table provides a schedule of building activities for a domestic building and the related duration in days.

ACTIVITY	DURATION [DAYS]
Site clearance	**3**
Setting out of building (concurrent with day THREE of site clearance)	**1**
Excavation of trenches (starts day after setting out ends)	**5**
Casting of concrete foundations and slab (starts day after excavation ends)	**2**
Superstructure brickwork (starts day after casting of concrete ends)	**10**
Erection of roof (starts day after superstructure brickwork ends)	**3**
Installation of doors and windows (starts day after erection of roof ends)	**5**
Plastering of internal walls and floor screeds (concurrent with day 1–3 of installation of windows and doors)	**3**
Plastering of external walls (starts after the fifth day of installation of windows and doors)	**5**
Painting of walls (starts day after external plastering ends)	**2**

Use the table above to complete the projection programme bar chart provided on the ANSWER SHEET.

The first row has been completed for you. **[10]**

TOTAL: 100

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ANSWER SHEET EXAMINATION NUMBER:

Detach and submit along with the ANSWER BOOK.

QUESTION 7: PROJECTION PROGRAMME BAR CHART

		Week 1					Week 2					Week 3					Week 4					Week 5					Week 6					Week 7				
Activity	Days	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
Site clearance	3																																			
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